

Software Requirements Specification (SRS)

Project: Munchkin.cro – Custom Crochet E-Commerce Platform

Group - 9:

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1. Introduction

1.1 Purpose

This document specifies the software requirements for “munchkin.cro”, a custom crochet e-commerce platform. It defines system functionality, constraints, and performance requirements for development and validation.

1.2 Intended Audience

- Developers
- QA/Test Engineers
- Project Managers
- Business Stakeholders

1.3 Intended Use

This SRS serves as a guideline for system design, development, testing, and deployment. It ensures all stakeholders share a common understanding of system behavior.

1.4 Product Scope

Munchkin.cro is a web-based e-commerce platform focused on custom crochet products. Unlike traditional online stores, this platform allows customers to:

- Customize product colors and designs
- Request fully personalized crochet items
- Communicate directly with the seller using chat and reference images

Objectives:

- Enable personalized crochet product creation
- Provide interactive customization tools
- Improve customer engagement through direct communication

1.5 Risk Definitions

- Miscommunication in customization requests
- Order mismatch due to unclear specifications
- Image upload failures
- Data/privacy issues in chat system
- System downtime affecting orders

2. Overall Description

2.1 User Classes and Characteristics

a. Customer

- Basic internet and browsing skills
- Interested in personalized crochet items

b. Admin (Seller)

- Manages products and customization requests
- Handles customer communication and order processing

2.2 User Needs

a. Customers

- Browse crochet items
- Customize colors and variations
- Upload reference images
- Chat with seller for custom requests

- Place and track orders

b. Admin

- Manage product listings
- Handle custom orders and chats
- Approve/reject customization requests
- Update order status

2.3 Operating Environment

- Frontend: Web browser (Chrome, Edge, Firefox, Safari)
- Backend: Server-side framework (Node.js)
- Database: MongoDB
- Platform: Responsive web application (mobile + desktop)

2.4 Constraints

- Requires stable internet connection
- Limited by hosting/server capacity
- Image upload size limitations
- Real-time chat requires additional backend support

2.5 Assumptions

- Users understand basic online shopping
- Customers can provide clear customization inputs
- Admin will manually handle complex customization requests

3. Requirements

3.1 Functional Requirements

FR1: User Registration

User can create an account using email and password.

FR2: User Login

User can log into the system securely.

FR3: Browse Products

User can view available crochet items with images and descriptions.

FR4: Product Customization

User can:

- Select color variations
- Modify available design attributes (e.g., size, pattern)

FR5: Custom Request Submission

User can submit fully custom requests by:

- Writing descriptions
- Uploading reference images

FR6: Image Upload

User can upload images for customization reference.

FR7: Chat System

User can chat with admin to:

- Discuss customization
- Share ideas or references

FR8: Add to Cart

User can add customized or standard products to cart.

FR9: Order Placement

User can place orders after confirming customization.

FR10: Order Tracking

User can track order status (pending, in progress, completed).

FR11: Admin Product Management

Admin can add/edit/delete crochet products.

FR12: Admin Custom Order Handling

Admin can:

- View customization requests
- Communicate with customers
- Approve or modify requests

FR13: Admin Chat Management

Admin can respond to customer messages in real-time or asynchronously.

FR14: Order Management

Admin can update order status and manage delivery workflow.

3.2 Non-Functional Requirements

Performance

- Page load time should be under 3 seconds
- Chat system should have minimal delay

Security

- Secure authentication system
- Encrypted user data and images

Usability

- Intuitive UI for customization
- Easy navigation for non-technical users

Reliability

System availability of at least 99%

Scalability

Ability to support increasing users and orders

Maintainability

Modular code structure for easy updates

Compatibility

Works across major browsers and devices

Conclusion

This SRS defines the requirements for munchkin.cro, a custom crochet e-commerce platform focused on personalization and customer interaction. It establishes a clear foundation for building a system that supports creative customization and efficient order management.